

A Survey on Mixture of Experts: Advancements, Challenges, and Future Directions

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Abstract—Mixture of Experts (MoE) is an advanced machine learning paradigm that enhances model scalability, efficiency, and specialization by dynamically selecting subsets of experts for each input. This survey provides a comprehensive overview of MoE, covering its fundamental principles, advantages, challenges, recent advancements, and future research directions. MoE architectures have been widely adopted in large-scale artificial intelligence systems, particularly in natural language processing and computer vision, due to their ability to optimize computational resources while maintaining high model capacity. Key challenges such as training instability, expert imbalance, computational overhead, and interpretability are discussed alongside potential mitigation strategies. Recent innovations, including hierarchical MoE, improved gating mechanisms, adaptive expert pruning, and MoE integration into multimodal learning, have significantly enhanced the framework's effectiveness. The survey also explores future research directions, focusing on improving training stability, fairness, continual learning, and resource efficiency. As MoE continues to evolve, it remains a promising approach for developing highly efficient and scalable AI systems.

Index Terms—Mixture of Experts, Deep Learning, Model Scalability, Expert Routing, Sparse Activation, Neural Networks, Large-Scale AI Models, Computational Efficiency, Gating Mechanisms, Interpretability, Load Balancing, Adaptive Learning

I. INTRODUCTION

The rapid advancements in deep learning have led to the development of increasingly large and complex models that achieve state-of-the-art performance across various domains, including natural language processing (NLP), computer vision, and speech recognition [1]. However, as model sizes grow, so do the computational costs and resource requirements, making it challenging to deploy such models efficiently [2]. To address this issue, researchers have explored different strategies to improve model efficiency while maintaining or even enhancing predictive performance [3]. One such approach is the Mixture of Experts (MoE), a modular neural network architecture that dynamically selects subsets of specialized models, called experts, to process each input [4]. Mixture of Experts was initially introduced as a method to enhance learning efficiency by dividing complex tasks into subtasks handled by different expert models. Each expert specializes in a specific aspect of the input space, and a gating mechanism is used to assign different inputs to the most relevant experts. This enables MoE to distribute computations efficiently, allowing only a subset of experts to be active at any given time, reducing the overall computational burden compared to dense models of equivalent size. In recent years, MoE has gained significant

traction, particularly in large-scale deep learning models [5]. The introduction of MoE-based architectures in deep learning has demonstrated the potential of this approach in scaling neural networks effectively [6]. These models use sparse activation mechanisms where only a fraction of experts is selected for each forward pass, significantly improving efficiency while maintaining strong performance on downstream tasks. Beyond efficiency, MoE provides several advantages, such as improved model interpretability, modular training, and enhanced adaptability to different types of data distributions [7]. By enabling different experts to focus on different types of inputs, MoE can achieve better generalization and robustness in complex real-world scenarios [8]. For instance, in NLP tasks, different experts can specialize in different linguistic patterns, domains, or contextual representations, leading to more effective language modeling. Despite its advantages, MoE also introduces several challenges that have been the subject of active research [9]. One major challenge is expert imbalance, where some experts are overutilized while others remain underutilized, leading to inefficient training and suboptimal generalization. Another challenge is training instability due to the discrete nature of expert selection, which can result in high variance in gradients and affect convergence [10]. Furthermore, communication overhead in distributed training settings can be significant, as routing decisions require efficient coordination across multiple processing units [11]. Several techniques, including improved routing mechanisms, load-balancing strategies, and better optimization techniques, have been proposed to mitigate these challenges. This survey aims to provide a comprehensive overview of Mixture of Experts, covering its theoretical foundations, recent advancements, and key challenges [12]. We systematically categorize existing MoE approaches, analyze their effectiveness across different application domains, and discuss emerging trends in the field [13]. By synthesizing recent developments, this survey serves as a reference for researchers and practitioners interested in leveraging MoE to build scalable, efficient, and high-performing machine learning systems [14]. The rest of this paper is organized as follows: Section II provides a theoretical background on MoE, including its mathematical formulation and fundamental principles [15]. Section III discusses various MoE architectures and their evolution in deep learning [16]. Section IV explores real-world applications of MoE across different domains. Section V highlights key challenges associated with MoE, along with proposed solutions. Finally, Section VI discusses future directions and open research problems in the field.

II. BACKGROUND

Mixture of Experts (MoE) is a neural network paradigm that enables dynamic routing of inputs to specialized subnetworks, known as experts [17]. This architecture is inspired by the principle of divide-and-conquer, where complex tasks are broken down into smaller, more manageable subproblems handled by specialized models [18]. A gating mechanism is responsible for selecting the most relevant experts for each input, ensuring that different parts of the model contribute effectively to the overall decision-making process.

A. Mathematical Formulation

An MoE model consists of a set of E expert networks, each parameterized by θ_e , where $e \in \{1, 2, \dots, E\}$. Given an input $x \in \mathbb{R}^d$, the MoE framework processes it as follows:

1. **Gating Function:** A gating network computes a probability distribution over the experts:

$$G(x) = \text{softmax}(W_g x + b_g)$$

where W_g and b_g are the parameters of the gating network [19]. The output $G(x) \in \mathbb{R}^E$ represents the probabilities assigned to each expert [20]. 2. **Expert Computation:** Each expert network processes the input independently, producing an output y_e :

$$y_e = f_e(x; \theta_e)$$

where $f_e(\cdot)$ is the function represented by the expert. 3. **Aggregation of Expert Outputs:** The final output is computed as a weighted sum of the expert outputs:

$$y = \sum_{e=1}^E G_e(x) y_e$$

where $G_e(x)$ represents the gating weight assigned to expert e . The gating network plays a crucial role in dynamically selecting the most relevant experts based on the input distribution, ensuring that computation is allocated efficiently [21].

B. Types of Mixture of Experts

MoE architectures can be categorized based on their gating and expert selection mechanisms:

- **Soft MoE:** The gating function assigns continuous weights to all experts, allowing multiple experts to contribute to the final output [22].
- **Hard MoE:** The gating function selects a fixed number of top experts (e.g., top- k selection), activating only a subset of experts per input.
- **Hierarchical MoE:** A multi-layer MoE structure where gating decisions occur at multiple levels, enabling a more fine-grained specialization of experts [23].

C. Comparison with Traditional Neural Networks

Unlike conventional neural networks where all model parameters are utilized for every input, MoE introduces sparsity by selectively activating only a subset of experts per inference step. This sparsity provides several benefits:

- **Scalability:** MoE allows training of very large models while keeping computational costs low by activating only a few experts at a time.
- **Modularity:** Experts can specialize in different aspects of the data distribution, improving interpretability and robustness [24].
- **Efficiency:** By distributing computations across experts, MoE reduces redundant processing and speeds up inference in large-scale models.

The effectiveness of MoE heavily depends on the quality of the gating function and the balance in expert utilization. Ensuring that experts are evenly utilized and avoiding overfitting to specific experts remain key research challenges. The next section will explore various MoE architectures, detailing their structural differences and practical implementations [25].

III. ARCHITECTURES OF MIXTURE OF EXPERTS

The design of Mixture of Experts (MoE) architectures has evolved significantly to improve efficiency, scalability, and adaptability [26]. Different architectural choices impact expert selection, training stability, and computational overhead [27]. This section explores the key variations in MoE architectures and their implementation strategies [28].

A. Basic MoE Architecture

A standard MoE model consists of three main components:

- **Expert Networks:** A set of independent subnetworks, each specializing in different aspects of the input space.
- **Gating Network:** A function that dynamically routes inputs to the most relevant experts [29].
- **Aggregation Mechanism:** A method for combining the outputs of the selected experts to produce a final prediction.

The gating network typically uses a softmax function to assign probability weights to each expert, ensuring that different inputs are processed by different subsets of experts. Depending on the routing strategy, MoE architectures can be classified into soft, hard, and hierarchical MoE models.

B. Soft MoE

Soft MoE assigns fractional weights to all experts, allowing each expert to contribute proportionally to the final output. The output is computed as a weighted sum:

$$y = \sum_{e=1}^E G_e(x) f_e(x),$$

where $G_e(x)$ represents the gating weight for expert e and $f_e(x)$ is the expert's output. **Advantages:**

- Ensures smooth gradient flow during training [30].
- Reduces abrupt switches between experts, improving stability.

Challenges:

- All experts are partially active, leading to higher computational costs.
- Less effective in enforcing strong specialization among experts.

C. Hard MoE

Hard MoE uses a top- k selection strategy, where only the k experts with the highest gating scores are activated for each input. The output is computed as:

$$y = \sum_{e \in \mathcal{S}_k} G_e(x) f_e(x),$$

where $\mathcal{S}_k$ is the set of the top- k selected experts. **Advantages:**

- Reduces computation by activating only a subset of experts [31].
- Encourages stronger specialization among experts.

Challenges:

- Introduces non-differentiability due to discrete selection, requiring specialized training techniques.
- Can lead to expert under-utilization and load imbalance [32].

D. Hierarchical MoE

Hierarchical MoE introduces multiple levels of expert selection, where a high-level gating network first selects groups of experts, and a lower-level gating function further refines the selection. This approach enables finer granularity in expert specialization. **Advantages:**

- Provides more structured specialization.
- Improves efficiency by introducing multiple levels of sparsity [33].

Challenges:

- More complex to train due to multi-stage routing [34].
- Increased risk of routing errors propagating through hierarchy [35].

E. Sparsely Activated MoE in Deep Learning

Recent advancements have integrated MoE into large-scale deep learning models, leveraging sparsity to scale neural networks efficiently. In transformer-based architectures, MoE layers are inserted in place of dense feedforward layers, allowing only a fraction of the experts to be activated per token or input. Key optimizations include:

- **Load-Balancing Loss:** Encourages equal utilization of experts by penalizing imbalanced routing [36].
- **Token-Level Routing:** Ensures that different tokens in a sequence can be routed to different experts.
- **Efficient Communication Mechanisms:** Reduces latency and synchronization overhead in distributed MoE training [37].

F. Comparison of MoE Architectures

The choice of MoE architecture depends on the trade-offs between computational efficiency, expert specialization, and training stability. While hard MoE is more efficient, it can be challenging to train, whereas soft MoE is computationally expensive but easier to optimize [38]. Hierarchical MoE provides a balance between the two but introduces additional complexity [39]. The next section explores the applications of Mixture of Experts across various domains, highlighting its effectiveness in real-world scenarios.

IV. APPLICATIONS OF MIXTURE OF EXPERTS

The Mixture of Experts (MoE) framework has been successfully applied to a wide range of domains, leveraging its ability to dynamically allocate computational resources and specialize in different data distributions. This section explores the key application areas where MoE has demonstrated significant improvements in efficiency, scalability, and performance [40].

A. Natural Language Processing

MoE has played a crucial role in scaling modern natural language processing (NLP) models. By incorporating sparsely activated expert layers into large transformer-based architectures, MoE enables efficient training and inference while maintaining or improving performance [41]. Key applications in NLP include:

- **Machine Translation:** MoE-based transformers have been used to enhance translation quality by enabling experts to specialize in different language pairs or linguistic structures.
- **Language Modeling:** Large-scale language models with MoE layers benefit from dynamic expert selection, improving performance on diverse linguistic inputs.
- **Text Generation:** MoE has been integrated into generative models to enhance fluency, coherence, and diversity in generated text.
- **Multilingual Models:** MoE enables efficient scaling of multilingual NLP models by allocating different experts to different languages, reducing redundancy in parameter usage.

B. Computer Vision

In computer vision, MoE has been applied to deep convolutional and transformer-based architectures to improve feature representation learning and computational efficiency [42]. Notable applications include:

- **Image Classification:** MoE enables adaptive feature extraction by routing different image regions to specialized experts, improving classification accuracy.
- **Object Detection and Segmentation:** MoE-based vision models enhance spatial awareness by allowing different experts to focus on distinct object types or scene contexts.
- **Few-Shot and Zero-Shot Learning:** MoE improves model adaptability to unseen categories by leveraging expert diversity.
- **Efficient Visual Processing:** By activating only a subset of experts, MoE reduces computational overhead in large-scale vision models [43].

C. Speech Recognition

Speech recognition models benefit from MoE by dynamically selecting experts based on acoustic variations, speaker identity, or linguistic context. Applications include:

- **Automatic Speech Recognition (ASR):** MoE enhances transcription accuracy by routing different audio signals to specialized phoneme or speaker experts.

Architecture	Computational Cost	Expert Specialization	Training Stability
Soft MoE	High	Weak	High
Hard MoE	Low (Sparse)	Strong	Low (Hard Selection)
Hierarchical MoE	Medium	Strongest	Medium

TABLE I
COMPARISON OF DIFFERENT MOE ARCHITECTURES.

- **Speech Synthesis:** MoE-based text-to-speech (TTS) models improve the naturalness and expressiveness of generated speech by allowing different experts to model different speech characteristics.
- **Speaker Adaptation:** MoE improves robustness to speaker variability by dynamically adjusting expert selection.

D. Reinforcement Learning

MoE has been employed in reinforcement learning (RL) to enhance exploration, policy learning, and decision-making in complex environments [44]. Key applications include:

- **Multi-Agent Systems:** MoE facilitates specialization among agents in multi-agent RL settings, improving coordination and efficiency [45].
- **Hierarchical RL:** MoE enables hierarchical decision-making by assigning different experts to different levels of task abstraction [46].
- **Adaptive Policy Learning:** MoE-based policies improve generalization by dynamically selecting specialized sub-policies for different task distributions [47].

E. Healthcare and Biomedical Applications

In healthcare and biomedical research, MoE has been applied to predictive modeling, medical imaging, and drug discovery. Applications include:

- **Disease Diagnosis:** MoE-based models improve diagnostic accuracy by allowing different experts to specialize in different medical conditions [48].
- **Medical Image Analysis:** Experts in MoE architectures can focus on different anatomical structures, improving segmentation and classification of medical images.
- **Drug Discovery:** MoE enables efficient exploration of molecular spaces by allocating different experts to different chemical properties or biological interactions [49].

F. Finance and Economic Modeling

MoE has been successfully utilized in financial forecasting, risk management, and economic modeling due to its ability to handle heterogeneous data distributions [50]. Applications include:

- **Stock Market Prediction:** MoE enhances forecasting accuracy by allowing experts to specialize in different market conditions and trends.
- **Risk Assessment:** Financial models leverage MoE to dynamically assess risk levels based on diverse economic indicators [51].
- **Fraud Detection:** MoE improves anomaly detection in financial transactions by enabling experts to focus on different fraud patterns.

G. Autonomous Systems and Robotics

Autonomous systems benefit from MoE by enabling modular decision-making and adaptive control strategies. Applications include:

- **Self-Driving Cars:** MoE enhances perception and decision-making in autonomous vehicles by allowing different experts to specialize in distinct driving scenarios.
- **Robotics Control:** MoE-based controllers improve adaptability in robotic manipulation and navigation tasks [52].
- **Human-Robot Interaction:** MoE facilitates personalized interaction models by dynamically selecting experts based on user behavior.

H. Comparison of MoE Performance Across Domains

The diverse applications of MoE demonstrate its versatility in handling complex tasks across multiple domains. By efficiently distributing computations and enabling expert specialization, MoE has proven to be a valuable approach in scaling AI models while maintaining high performance [53]. The next section explores key challenges associated with MoE, including training instability, expert imbalance, and computational overhead, along with potential solutions.

V. CHALLENGES AND LIMITATIONS

Despite the numerous advantages of Mixture of Experts (MoE) in scalability, efficiency, and specialization, several challenges must be addressed to ensure stable and effective deployment. These challenges include training instability, expert imbalance, computational overhead, and interpretability issues [54].

A. Training Instability

One of the primary challenges in MoE models is training instability [55]. The presence of a gating network that dynamically selects experts can introduce instability in the learning process due to the following reasons:

- **Sparse Activation:** Hard MoE models activate only a subset of experts per input, leading to uneven gradient updates, which can destabilize training.
- **Gating Network Sensitivity:** The gating function's output determines expert selection, making training highly sensitive to small weight changes.
- **Exploding and Vanishing Gradients:** Backpropagation through sparse expert selection can lead to exploding or vanishing gradients, impacting model convergence.

Potential Solutions:

- Regularization techniques such as entropy loss on gating outputs can encourage more uniform expert utilization [56].

Application Domain	Primary Benefit	Key Challenge	Example Use Case
NLP	Scalability	Expert imbalance	Machine translation
Computer Vision	Adaptive feature learning	Training stability	Image classification
Speech Recognition	Speaker adaptation	Routing efficiency	ASR systems
Reinforcement Learning	Policy specialization	Sparse rewards	Multi-agent RL
Healthcare	Medical expertise specialization	Data scarcity	Disease diagnosis
Finance	Market adaptation	Model interpretability	Stock prediction
Autonomous Systems	Modular decision-making	Real-time processing	Self-driving cars

TABLE II
COMPARISON OF MoE APPLICATIONS ACROSS DIFFERENT DOMAINS.

- Gradient clipping and careful weight initialization can mitigate extreme gradient updates.
- Warm-up strategies for gating parameters can improve convergence stability.

B. Expert Imbalance and Load Balancing

A significant issue in MoE models is expert imbalance, where certain experts receive a disproportionate share of inputs while others remain underutilized [57]. This imbalance can lead to:

- Poor generalization due to overfitting of frequently selected experts.
- Inefficient resource utilization, as underused experts contribute little to the model's overall performance [58].
- Slower convergence, as only a subset of experts receives meaningful gradient updates.

Potential Solutions:

- **Load Balancing Loss:** Adding a regularization term that penalizes extreme expert selection biases [59].
- **Noisy Gating:** Injecting noise into the gating network during training to encourage more diverse expert usage [60].
- **Temperature Scaling in Softmax:** Adjusting the temperature parameter of the gating softmax function to control sparsity.

C. Computational Overhead

Efficient Mixture of Experts (MoE) models optimize computational resources by selectively activating only a subset of experts per input, reducing overall computational overhead while maintaining high performance. Several techniques enhance MoE efficiency, including pruning, decomposition, quantization, and memory optimization. Pruning strategies, such as static and dynamic pruning, remove redundant experts or parameters to improve expert utilization and reduce memory consumption. Decomposition methods, including low-rank factorization and singular value decomposition, compress expert weight matrices while preserving expressiveness. Quantization techniques, such as post-training quantization (PTQ) and quantization-aware training (QAT), lower model precision to decrease storage requirements and accelerate inference. Additionally, memory and communication optimizations, such as expert parallelism, sparse expert updates, gradient compression, and activation checkpointing, further reduce training and deployment costs. These techniques collectively improve the

scalability and efficiency of MoE, making it suitable for large-scale AI applications and deployment on resource-constrained devices [4], [61], [62], [63].

Although MoE models improve efficiency by activating only a subset of experts, they introduce significant computational overhead in the following areas:

- **Increased Memory Requirements:** Large MoE models require substantial memory for storing multiple expert parameters.
- **Inefficient Batch Processing:** Since different inputs may be routed to different experts, batching operations become less efficient [62].
- **High Communication Costs in Distributed Training:** In multi-GPU or multi-node training, expert routing introduces significant inter-device communication overhead [64].

Potential Solutions:

- **Expert Pruning:** Removing redundant experts based on activation frequency to reduce model size [65], [41].
- **Efficient Parallelization:** Optimizing expert placement across GPUs to minimize communication latency.
- **Sparse Matrix Operations:** Leveraging specialized hardware accelerations (e.g., TPU optimizations) for sparse computation.

D. Interpretability and Explainability

MoE models introduce additional complexity in interpretability, as decision-making is distributed across multiple experts. Understanding which expert contributes to a given decision can be challenging due to:

- Lack of clear attribution of decision-making to individual experts [66].
- Difficulty in visualizing expert specialization [67].
- Challenges in debugging and auditing MoE-based predictions [68].

Potential Solutions:

- **Expert Attention Visualization:** Analyzing the gating network's weight distributions to understand expert selection patterns.
- **Explainable AI (XAI) Techniques:** Applying SHAP values, feature attribution methods, or attention-based interpretability tools to MoE outputs.
- **Constraint-Based Training:** Encouraging domain-specific expert specialization using structured constraints [69].

E. Overfitting and Generalization

MoE models, particularly when overparameterized, risk overfitting on training data due to their ability to memorize patterns rather than generalizing. Overfitting can be exacerbated by:

- **Sparse Expert Utilization:** If only a few experts dominate predictions, the model may fail to generalize to unseen data [70].
- **High Model Capacity:** Large MoE models with numerous experts may capture dataset-specific noise [71].

Potential Solutions:

- **Dropout for Experts:** Randomly deactivating experts during training to encourage robustness [72].
- **Data Augmentation:** Expanding the dataset to ensure experts learn diverse patterns [73].
- **Weight Sharing:** Allowing partial parameter sharing among experts to reduce overfitting [74].

F. Ethical Considerations and Fairness

The use of MoE in critical decision-making systems introduces ethical concerns, particularly regarding fairness and bias:

- **Bias in Expert Selection:** If the gating function is biased, certain subgroups in the data may consistently receive different model responses [75].
- **Lack of Transparency:** MoE models can be harder to audit for fairness due to dynamic expert routing.
- **Data Privacy Risks:** Large-scale MoE models trained on diverse datasets may inadvertently learn and amplify sensitive patterns.

Potential Solutions:

- **Fairness Constraints in Gating Networks:** Encouraging balanced expert selection across demographic groups [76].
- **Bias Audits and Debiasing Strategies:** Regularly evaluating expert utilization statistics to detect disparities [77].
- **Privacy-Preserving MoE:** Employing federated learning or differential privacy techniques in MoE training.

G. Summary of Challenges and Mitigation Strategies

Addressing these challenges is crucial for the widespread adoption of MoE models in real-world applications. Ongoing research aims to refine expert routing mechanisms, improve training stability, and enhance interpretability while maintaining computational efficiency. The next section will explore recent advancements and future directions in MoE research, highlighting innovations aimed at overcoming these limitations [78].

VI. RECENT ADVANCEMENTS AND FUTURE DIRECTIONS

Mixture of Experts (MoE) has seen rapid advancements in recent years, driven by the need for scalable and efficient deep learning architectures. This section highlights recent innovations in MoE models and explores potential future research directions aimed at addressing existing challenges and expanding the applicability of MoE.

A. Advancements in MoE Architectures

Several architectural improvements have been introduced to enhance the performance and efficiency of MoE models:

- **Improved Gating Mechanisms:** Recent research has focused on more efficient gating functions to address expert imbalance and improve routing decisions. Techniques such as learnable temperature scaling, auxiliary load-balancing losses, and soft gating functions help mitigate expert underutilization [79].
- **Hierarchical MoE:** Multi-level MoE architectures have been proposed, where experts are organized hierarchically, allowing for finer-grained specialization and improved computational efficiency.
- **Conditional Computation:** Advances in sparse MoE models leverage conditional computation techniques to activate different subsets of experts dynamically based on input complexity, reducing unnecessary computation.
- **MoE in Vision Transformers (ViTs):** MoE has been integrated into transformer-based computer vision models to enhance scalability while maintaining accuracy, particularly in large-scale image classification and object detection tasks [80].
- **Efficient MoE for Edge Computing:** Efforts have been made to develop lightweight MoE models that can be deployed on edge devices, reducing memory and energy consumption while maintaining high performance [81].

B. Optimization and Training Strategies

Recent advancements in optimization techniques have contributed to more stable and efficient training of MoE models:

- **Gradient-Based Load Balancing:** Novel regularization techniques have been introduced to dynamically balance expert utilization during training, reducing the risk of expert collapse [82].
- **Sparse Expert Updates:** Methods that update only a subset of experts per training step help in reducing memory consumption and improving convergence speed.
- **Adaptive Expert Pruning:** Dynamic expert selection mechanisms allow pruning underutilized experts during training, leading to more efficient models.
- **Self-Supervised Learning with MoE:** Self-supervised approaches have been integrated with MoE architectures to improve learning efficiency, especially in unsupervised and semi-supervised tasks [83].

C. Scaling MoE for Large-Scale AI Models

MoE has been instrumental in the development of extremely large-scale AI models, particularly in NLP and multimodal learning:

- **MoE in Large-Scale Language Models:** MoE architectures have enabled training billion- and trillion-parameter language models while maintaining manageable computational costs [84].
- **MoE for Multimodal Learning:** Recent research has explored integrating MoE into models that handle multiple

Challenge	Impact	Proposed Solution
Training Instability	Poor convergence	Gradient clipping, warm-up strategies
Expert Imbalance	Underutilization of experts	Load-balancing loss, noisy gating
Computational Overhead	High memory usage	Expert pruning, parallelization
Interpretability Issues	Lack of transparency	Expert attention analysis, explainable AI
Overfitting	Poor generalization	Dropout for experts, weight sharing
Ethical Concerns	Bias, fairness	Fairness-aware gating, bias audits

TABLE III
CHALLENGES IN MoE MODELS AND THEIR POTENTIAL SOLUTIONS.

data modalities (e.g., text, image, audio), enabling cross-modal specialization [85].

- **Distributed MoE Training:** Advances in distributed training techniques, such as expert parallelism and model partitioning, have helped scale MoE models across thousands of GPUs [86].

D. Interpretable and Explainable MoE Models

Improving the interpretability of MoE models remains an active research area:

- **Explainable Routing Mechanisms:** Methods have been developed to make expert selection more transparent, including attention-based gating visualizations [87].
- **Post-Hoc Interpretability:** Techniques such as SHAP values and feature attribution methods have been adapted for MoE architectures to analyze expert contributions.
- **Domain-Specific Specialization:** Research into constraining experts based on domain knowledge has improved model interpretability and robustness [88].

E. Future Research Directions

Despite recent progress, several open challenges remain, presenting opportunities for future research:

- **Reducing Computational Overhead:** Developing more efficient MoE architectures that reduce communication overhead in distributed settings while maintaining performance.
- **Robustness and Generalization:** Improving generalization across tasks and reducing expert overfitting remains a key challenge [89].
- **Fairness and Bias Mitigation:** Ensuring equitable expert selection across demographic groups and minimizing bias in decision-making processes.
- **Integration with Neuroscientific and Cognitive Models:** Drawing inspiration from biological neural systems to design more adaptive and efficient expert selection mechanisms.
- **MoE for Continual Learning:** Investigating MoE's potential in lifelong learning scenarios where models need to adapt continuously to new tasks [90].
- **Hybrid MoE Architectures:** Combining MoE with other architectures, such as reinforcement learning or meta-learning, to create more adaptive AI systems.

F. Summary of Recent Trends and Future Directions

The continued evolution of MoE models promises to enhance their applicability across diverse domains [91]. As research progresses, advancements in efficiency, interpretability,

and fairness will play a crucial role in ensuring the widespread adoption of MoE-based architectures [92]. The next section will provide a conclusion summarizing the key insights from this survey and highlighting the future impact of MoE in artificial intelligence [93].

VII. CONCLUSION

Mixture of Experts (MoE) has emerged as a powerful paradigm for scaling deep learning models while maintaining computational efficiency [94]. By dynamically selecting subsets of experts for each input, MoE enables specialization, improves efficiency, and enhances the scalability of large-scale models [95]. This survey has explored the fundamental principles of MoE, its advantages, challenges, recent advancements, and future research directions. The survey highlights several key findings:

- MoE architectures effectively improve computational efficiency by leveraging sparse activation, enabling the development of large-scale AI models without a proportional increase in computational cost.
- Advanced gating mechanisms and load-balancing techniques have significantly mitigated expert imbalance issues, ensuring better resource utilization [96].
- Recent developments in hierarchical MoE, conditional computation, and expert pruning have further enhanced the efficiency and flexibility of MoE models [97].
- Challenges such as training instability, expert underutilization, and interpretability remain active research areas, requiring innovative solutions to ensure robust and fair deployment [98].
- MoE has demonstrated significant potential in diverse domains, including natural language processing, computer vision, multimodal learning, and edge computing [99].
- Future research directions focus on improving training stability, reducing computational overhead, enhancing fairness and bias mitigation, and integrating MoE with other learning paradigms such as continual learning and reinforcement learning.

Despite its challenges, MoE continues to shape the future of AI by providing a scalable and adaptive learning framework. The ability to efficiently allocate computational resources while maintaining high model capacity makes MoE an attractive approach for next-generation AI systems [100]. As research in MoE progresses, further innovations in expert routing, optimization techniques, and interpretability will drive its adoption in real-world applications.

In conclusion, MoE represents a crucial step toward building more efficient and scalable AI systems. Addressing existing

Research Area	Recent Advancements	Future Directions
Architecture Design	Hierarchical MoE, Vision MoE	More efficient MoE for edge devices
Training Optimization	Load balancing, adaptive pruning	Scalable expert routing strategies
Scaling Large Models	MoE for billion-parameter models	Efficient distributed MoE training
Interpretability	Expert selection visualization	Explainable MoE routing mechanisms
Fairness	Bias audits in MoE	Ethical MoE deployment frameworks
Continual Learning	MoE for dynamic tasks	Lifelong learning with MoE

TABLE IV
RECENT ADVANCEMENTS AND FUTURE RESEARCH DIRECTIONS IN MoE.

limitations and leveraging advancements in deep learning architectures will ensure that MoE continues to play a transformative role in the development of intelligent models across various domains.

REFERENCES

- [1] B. Mustafa, C. Riquelme, J. Puigcerver, R. Jenatton, and N. Houlsby, "Multimodal contrastive learning with limoe: the language-image mixture of experts," *Advances in Neural Information Processing Systems*, vol. 35, pp. 9564–9576, 2022.
- [2] B. Shabbaba and R. Neal, "Nonlinear models using Dirichlet process mixtures," *Journal of Machine Learning Research*, vol. 10, no. 8, 2009.
- [3] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," *Advances in neural information processing systems*, vol. 30, 2017.
- [4] S. Li, F. Xue, C. Baranwal, Y. Li, and Y. You, "Sequence parallelism: Long sequence training from system perspective," *arXiv preprint arXiv:2105.13120*, 2021.
- [5] Y. Zhu, N. Wichers, C.-C. Lin, X. Wang, T. Chen, L. Shu, H. Lu, C. Liu, L. Luo, J. Chen *et al.*, "Sira: Sparse mixture of low rank adaptation," *arXiv preprint arXiv:2311.09179*, 2023.
- [6] C. Shi, C. Yang, X. Zhu, J. Wang, T. Wu, S. Li, D. Cai, Y. Yang, and Y. Meng, "Unchosen experts can contribute too: Unleashing moe models' power by self-contrast," *arXiv preprint arXiv:2405.14507*, 2024.
- [7] S. He, D. Dong, L. Ding, and A. Li, "Demystifying the compression of mixture-of-experts through a unified framework," *arXiv preprint arXiv:2406.02500*, 2024.
- [8] Z. Zhang, Z. Zeng, Y. Lin, C. Xiao, X. Wang, X. Han, Z. Liu, R. Xie, M. Sun, and J. Zhou, "Emergent modularity in pre-trained transformers," in *Findings of the Association for Computational Linguistics: ACL 2023*, 2023, pp. 4066–4083.
- [9] Y. Du, S. Zhao, D. Zhao, M. Ma, Y. Chen, L. Huo, Q. Yang, D. Xu, and B. Qin, "Mogu: A framework for enhancing safety of open-sourced llms while preserving their usability," *arXiv preprint arXiv:2405.14488*, 2024.
- [10] G. Lample, A. Sablayrolles, M. Ranzato, L. Denoyer, and H. Jégou, "Large memory layers with product keys," *Advances in Neural Information Processing Systems*, vol. 32, 2019.
- [11] J. Yao, Q. Anthony, A. Shafi, H. Subramoni *et al.*, "Exploiting Inter-Layer Expert Affinity for Accelerating Mixture-of-Experts Model Inference," *arXiv preprint arXiv:2401.08383*, 2024.
- [12] V. A. Korthikanti, J. Casper, S. Lym, L. McAfee, M. Andersch, M. Shoenybi, and B. Catanzaro, "Reducing activation recomputation in large transformer models," *Proceedings of Machine Learning and Systems*, vol. 5, 2023.
- [13] W. Kwon, Z. Li, S. Zhuang, Y. Sheng, L. Zheng, C. H. Yu, J. Gonzalez, H. Zhang, and I. Stoica, "Efficient memory management for large language model serving with pagedattention," in *Proceedings of the 29th Symposium on Operating Systems Principles*, 2023, pp. 611–626.
- [14] D. Dua, S. Bhosale, V. Goswami, J. Cross, M. Lewis, and A. Fan, "Tricks for Training Sparse Translation Models," in *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 2022, pp. 3340–3345.
- [15] Z. Chen, Y. Shen, M. Ding, Z. Chen, H. Zhao, E. G. Learned-Miller, and C. Gan, "Mod-squad: Designing mixtures of experts as modular multi-task learners," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 11 828–11 837.
- [16] N. Du, Y. Huang, A. M. Dai, S. Tong, D. Lepikhin, Y. Xu, M. Krikun, Y. Zhou, A. W. Yu, O. Firat *et al.*, "Glam: Efficient scaling of language models with mixture-of-experts," in *International Conference on Machine Learning*. PMLR, 2022, pp. 5547–5569.
- [17] B. McKinzie, Z. Gan, J.-P. Fauconnier, S. Dodge, B. Zhang, P. Dufer, D. Shah, X. Du, F. Peng, F. Weers *et al.*, "Mml: Methods, analysis & insights from multimodal llm pre-training," *arXiv preprint arXiv:2403.09611*, 2024.
- [18] O. Ostapenko, L. Caccia, Z. Su, N. Le Roux, L. Charlin, and A. Sordani, "A Case Study of Instruction Tuning with Mixture of Parameter-Efficient Experts," in *NeurIPS 2023 Workshop on Instruction Tuning and Instruction Following*, 2023.
- [19] M. N. R. Chowdhury, S. Zhang, M. Wang, S. Liu, and P.-Y. Chen, "Patch-level routing in mixture-of-experts is provably sample-efficient for convolutional neural networks," in *International Conference on Machine Learning*. PMLR, 2023, pp. 6074–6114.
- [20] R. Aljundi, P. Chakravarty, and T. Tuytelaars, "Expert gate: Lifelong learning with a network of experts," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2017, pp. 3366–3375.
- [21] K. Cobbe, V. Kosaraju, M. Bavarian, M. Chen, H. Jun, L. Kaiser, M. Plappert, J. Tworek, J. Hilton, R. Nakano *et al.*, "Training verifiers to solve math word problems," *arXiv preprint arXiv:2110.14168*, 2021.
- [22] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "Bert: Pre-training of deep bidirectional transformers for language understanding," *arXiv preprint arXiv:1810.04805*, 2018.
- [23] J. Wei, X. Wang, D. Schuurmans, M. Bosma, F. Xia, E. Chi, Q. V. Le, D. Zhou *et al.*, "Chain-of-thought prompting elicits reasoning in large language models," *Advances in Neural Information Processing Systems*, vol. 35, pp. 24 824–24 837, 2022.
- [24] B. Lin, Z. Tang, Y. Ye, J. Cui, B. Zhu, P. Jin, J. Zhang, M. Ning, and L. Yuan, "Moe-llava: Mixture of experts for large vision-language models," *arXiv preprint arXiv:2401.15947*, 2024.
- [25] L. Zhou, H. Palangi, L. Zhang, H. Hu, J. Corso, and J. Gao, "Unified vision-language pre-training for image captioning and vqa," in *Proceedings of the AAAI conference on artificial intelligence*, vol. 34, no. 07, 2020, pp. 13 041–13 049.
- [26] Z. Han, C. Gao, J. Liu, S. Q. Zhang *et al.*, "Parameter-efficient fine-tuning for large models: A comprehensive survey," *arXiv preprint arXiv:2403.14608*, 2024.
- [27] X. Ren, P. Zhou, X. Meng, X. Huang, Y. Wang, W. Wang, P. Li, X. Zhang, A. Podolskiy, G. Arshinov *et al.*, "Pangu-Σ: Towards trillion parameter language model with sparse heterogeneous computing," *arXiv preprint arXiv:2303.10845*, 2023.
- [28] J. Jiang, P. Zhang, Y. Luo, C. Li, J. B. Kim, K. Zhang, S. Wang, X. Xie, and S. Kim, "AdaMCT: adaptive mixture of CNN-transformer for sequential recommendation," in *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management*, 2023, pp. 976–986.
- [29] DeepSeek-AI, "DeepSeek-V2: A Strong, Economical, and Efficient Mixture-of-Experts Language Model," 2024.
- [30] R. Yi, L. Guo, S. Wei, A. Zhou, S. Wang, and M. Xu, "Edgemoe: Fast on-device inference of moe-based large language models," *arXiv preprint arXiv:2308.14352*, 2023.
- [31] S. He, L. Ding, D. Dong, B. Liu, F. Yu, and D. Tao, "Pad-net: An efficient framework for dynamic networks," in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2023, pp. 14 354–14 366.
- [32] J. Wei, Y. Tay, R. Bommasani, C. Raffel, B. Zoph, S. Borgeaud, D. Yogatama, M. Bosma, D. Zhou, D. Metzler *et al.*, "Emergent abilities of large language models," *arXiv preprint arXiv:2206.07682*, 2022.
- [33] X. Nie, P. Zhao, X. Miao, T. Zhao, and B. Cui, "HetuMoE: An efficient trillion-scale mixture-of-expert distributed training system," *arXiv preprint arXiv:2203.14685*, 2022.
- [34] X. Glorot, A. Bordes, and Y. Bengio, "Deep sparse rectifier neural networks," in *Proceedings of the fourteenth international conference on*

- artificial intelligence and statistics*. JMLR Workshop and Conference Proceedings, 2011, pp. 315–323.
- [35] J. Su, M. Ahmed, Y. Lu, S. Pan, W. Bo, and Y. Liu, “Roformer: Enhanced transformer with rotary position embedding,” *Neurocomputing*, vol. 568, p. 127063, 2024.
- [36] S. Singh, O. Ruwase, A. A. Awan, S. Rajbhandari, Y. He, and A. Bhatlele, “A Hybrid Tensor-Expert-Data Parallelism Approach to Optimize Mixture-of-Experts Training,” in *Proceedings of the 37th International Conference on Supercomputing*, 2023, pp. 203–214.
- [37] R. Zhang, J. Han, C. Liu, A. Zhou, P. Lu, H. Li, P. Gao, and Y. Qiao, “LLaMA-Adapter: Efficient Fine-tuning of Large Language Models with Zero-initialized Attention,” in *The Twelfth International Conference on Learning Representations*, 2024. [Online]. Available: <https://openreview.net/forum?id=d4UiXAHN2W>
- [38] H. Liu, D. Tam, M. Muqeeth, J. Mohta, T. Huang, M. Bansal, and C. A. Raffel, “Few-shot parameter-efficient fine-tuning is better and cheaper than in-context learning,” *Advances in Neural Information Processing Systems*, vol. 35, pp. 1950–1965, 2022.
- [39] S. Diao, T. Xu, R. Xu, J. Wang, and T. Zhang, “Mixture-of-Domain-Adapters: Decoupling and Injecting Domain Knowledge to Pre-trained Language Models’ Memories,” in *The 61st Annual Meeting Of The Association For Computational Linguistics*, 2023.
- [40] S. Dou, E. Zhou, Y. Liu, S. Gao, J. Zhao, W. Shen, Y. Zhou, Z. Xi, X. Wang, X. Fan *et al.*, “The Art of Balancing: Revolutionizing Mixture of Experts for Maintaining World Knowledge in Language Model Alignment,” *arXiv preprint arXiv:2312.09979*, 2023.
- [41] R. Hwang, J. Wei, S. Cao, C. Hwang, X. Tang, T. Cao, M. Yang, and M. Rhu, “Pre-gated MoE: An Algorithm-System Co-Design for Fast and Scalable Mixture-of-Expert Inference,” *arXiv preprint arXiv:2308.12066*, 2023.
- [42] S. A. R. Team, “Snowflake Arctic: The Best LLM for Enterprise AI — Efficiently Intelligent, Truly Open,” April 2024. [Online]. Available: <https://www.snowflake.com/blog/arctic-open-efficient-foundation-language-models-snowflake/>
- [43] N. Shazeer, A. Mirhoseini, K. Maziarz, A. Davis, Q. Le, G. Hinton, and J. Dean, “Outrageously large neural networks: The sparsely-gated mixture-of-experts layer,” *arXiv preprint arXiv:1701.06538*, 2017.
- [44] C. Rosenbaum, I. Cases, M. Riemer, and T. Klinger, “Routing networks and the challenges of modular and compositional computation,” *arXiv preprint arXiv:1904.12774*, 2019.
- [45] S. Shi, X. Pan, X. Chu, and B. Li, “Pipemoe: Accelerating mixture-of-experts through adaptive pipelining,” in *IEEE INFOCOM 2023-IEEE Conference on Computer Communications*. IEEE, 2023, pp. 1–10.
- [46] J. Kirkpatrick, R. Pascanu, N. Rabinowitz, J. Veness, G. Desjardins, A. A. Rusu, K. Milan, J. Quan, T. Ramalho, A. Grabska-Barwinska *et al.*, “Overcoming catastrophic forgetting in neural networks,” *Proceedings of the national academy of sciences*, vol. 114, no. 13, pp. 3521–3526, 2017.
- [47] C. Jiang, Y. Tian, Z. Jia, S. Zheng, C. Wu, and Y. Wang, “Lancet: Accelerating Mixture-of-Experts Training via Whole Graph Computation-Communication Overlapping,” *arXiv preprint arXiv:2404.19429*, 2024.
- [48] Databricks, “Introducing DBRX: A New State-of-the-Art Open LLM,” March 2024. [Online]. Available: <https://www.databricks.com/blog/introducing-dbrx-new-state-art-open-llm>
- [49] R. J. Williams, “Simple statistical gradient-following algorithms for connectionist reinforcement learning,” *Machine learning*, vol. 8, pp. 229–256, 1992.
- [50] A. Aghajanyan, S. Gupta, and L. Zettlemoyer, “Intrinsic Dimensionality Explains the Effectiveness of Language Model Fine-Tuning,” in *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, 2021, pp. 7319–7328.
- [51] N. Shazeer, “Glu variants improve transformer,” *arXiv preprint arXiv:2002.05202*, 2020.
- [52] O. Lieber, B. Lenz, H. Bata, G. Cohen, J. Osin, I. Dalmedigos, E. Safahi, S. Meirum, Y. Belinkov, S. Shalev-Shwartz *et al.*, “Jamba: A hybrid transformer-mamba language model,” *arXiv preprint arXiv:2403.19887*, 2024.
- [53] D. Lepikhin, H. Lee, Y. Xu, D. Chen, O. Firat, Y. Huang, M. Krikun, N. Shazeer, and Z. Chen, “Gshard: Scaling giant models with conditional computation and automatic sharding,” *arXiv preprint arXiv:2006.16668*, 2020.
- [54] A. Almahairi, N. Ballas, T. Cooijmans, Y. Zheng, H. Larochelle, and A. Courville, “Dynamic capacity networks,” in *International Conference on Machine Learning*. PMLR, 2016, pp. 2549–2558.
- [55] S. Shi, X. Pan, Q. Wang, C. Liu, X. Ren, Z. Hu, Y. Yang, B. Li, and X. Chu, “Schemoe: An extensible mixture-of-experts distributed training system with tasks scheduling,” in *Proceedings of the Nineteenth European Conference on Computer Systems*, 2024, pp. 236–249.
- [56] J. Ren, S. Rajbhandari, R. Y. Aminabadi, O. Ruwase, S. Yang, M. Zhang, D. Li, and Y. He, “{Zero-offload}: Democratizing {billion-scale} model training,” in *2021 USENIX Annual Technical Conference (USENIX ATC 21)*, 2021, pp. 551–564.
- [57] J. He, J. Qiu, A. Zeng, Z. Yang, J. Zhai, and J. Tang, “Fastmoe: A fast mixture-of-expert training system,” *arXiv preprint arXiv:2103.13262*, 2021.
- [58] H. Hazimeh, Z. Zhao, A. Chowdhery, M. Sathiamoorthy, Y. Chen, R. Mazumder, L. Hong, and E. Chi, “Dselect-k: Differentiable selection in the mixture of experts with applications to multi-task learning,” *Advances in Neural Information Processing Systems*, vol. 34, pp. 29 335–29 347, 2021.
- [59] Q. Team, “Introducing Qwen1.5,” February 2024. [Online]. Available: <https://qwenlm.github.io/blog/qwen1.5/>
- [60] S. Roller, S. Sukhbaatar, J. Weston *et al.*, “Hash layers for large sparse models,” *Advances in Neural Information Processing Systems*, vol. 34, pp. 17 555–17 566, 2021.
- [61] C. Riquelme, J. Puigcerver, B. Mustafa, M. Neumann, R. Jenatton, A. Susano Pinto, D. Keysers, and N. Houlsby, “Scaling vision with sparse mixture of experts,” *Advances in Neural Information Processing Systems*, vol. 34, pp. 8583–8595, 2021.
- [62] M. Muqeeth, H. Liu, and C. Raffel, “Soft merging of experts with adaptive routing,” *arXiv preprint arXiv:2306.03745*, 2023.
- [63] Y. Zniyed, T. P. Nguyen *et al.*, “Enhanced network compression through tensor decompositions and pruning,” *IEEE Transactions on Neural Networks and Learning Systems*, 2024.
- [64] C. Huang, Q. Liu, B. Y. Lin, T. Pang, C. Du, and M. Lin, “Lorahub: Efficient cross-task generalization via dynamic lora composition,” *arXiv preprint arXiv:2307.13269*, 2023.
- [65] Y. Zniyed, T. P. Nguyen *et al.*, “Efficient tensor decomposition-based filter pruning,” *Neural Networks*, vol. 178, p. 106393, 2024.
- [66] S. Dou, E. Zhou, Y. Liu, S. Gao, J. Zhao, W. Shen, Y. Zhou, Z. Xi, X. Wang, X. Fan *et al.*, “Loramoe: Revolutionizing mixture of experts for maintaining world knowledge in language model alignment,” *arXiv preprint arXiv:2312.09979*, 2023.
- [67] D. Xiao, H. Zhang, Y. Li, Y. Sun, H. Tian, H. Wu, and H. Wang, “ERNIE-GEN: an enhanced multi-flow pre-training and fine-tuning framework for natural language generation,” in *Proceedings of the Twenty-Ninth International Conference on International Joint Conferences on Artificial Intelligence*, 2021, pp. 3997–4003.
- [68] D. H. Dat, P. Y. Mao, T. H. Nguyen, W. Buntine, and M. Bennamoun, “HOMOE: A Memory-Based and Composition-Aware Framework for Zero-Shot Learning with Hopfield Network and Soft Mixture of Experts,” *arXiv preprint arXiv:2311.14747*, 2023.
- [69] S. Wu, J. Luo, X. Chen, L. Li, X. Zhao, T. Yu, C. Wang, Y. Wang, F. Wang, W. Qiao *et al.*, “Yuan 2.0-M32: Mixture of Experts with Attention Router,” *arXiv preprint arXiv:2405.17976*, 2024.
- [70] K. Punniyamurthy, K. Hamidouche, and B. M. Beckmann, “Optimizing distributed ml communication with fused computation-collective operations,” *arXiv preprint arXiv:2305.06942*, 2023.
- [71] H. Tang, J. Liu, M. Zhao, and X. Gong, “Progressive layered extraction (ple): A novel multi-task learning (mtl) model for personalized recommendations,” in *Proceedings of the 14th ACM Conference on Recommender Systems*, 2020, pp. 269–278.
- [72] Y. Huang, Y. Cheng, A. Bapna, O. Firat, D. Chen, M. Chen, H. Lee, J. Ngiam, Q. V. Le, Y. Wu *et al.*, “Gpipe: Efficient training of giant neural networks using pipeline parallelism,” *Advances in neural information processing systems*, vol. 32, 2019.
- [73] T. Gale, D. Narayanan, C. Young, and M. Zaharia, “Megablocks: Efficient sparse training with mixture-of-experts,” *Proceedings of Machine Learning and Systems*, vol. 5, 2023.
- [74] Y. Zhou, T. Lei, H. Liu, N. Du, Y. Huang, V. Zhao, A. M. Dai, Q. V. Le, J. Laudon *et al.*, “Mixture-of-experts with expert choice routing,” *Advances in Neural Information Processing Systems*, vol. 35, pp. 7103–7114, 2022.
- [75] T. Baltrušaitis, C. Ahuja, and L.-P. Morency, “Multimodal machine learning: A survey and taxonomy,” *IEEE transactions on pattern analysis and machine intelligence*, vol. 41, no. 2, pp. 423–443, 2018.
- [76] A. Clark, D. de Las Casas, A. Guy, A. Mensch, M. Paganini, J. Hoffmann, B. Damoc, B. Hechtman, T. Cai, S. Borgeaud *et al.*, “Unified scaling laws for routed language models,” in *International conference on machine learning*. PMLR, 2022, pp. 4057–4086.

- [77] L.-M. Team, “LLaMA-MoE: Building Mixture-of-Experts from LLaMA with Continual Pre-training,” Dec 2023. [Online]. Available: <https://github.com/pjlab-sys4nlp/llama-moe>
- [78] T. Luo, J. Lei, F. Lei, W. Liu, S. He, J. Zhao, and K. Liu, “Moelora: Contrastive learning guided mixture of experts on parameter-efficient fine-tuning for large language models,” *arXiv preprint arXiv:2402.12851*, 2024.
- [79] C. Chen, M. Li, Z. Wu, D. Yu, and C. Yang, “Ta-moe: Topology-aware large scale mixture-of-expert training,” *Advances in Neural Information Processing Systems*, vol. 35, pp. 22 173–22 186, 2022.
- [80] B. Zhang and R. Sennrich, “Root mean square layer normalization,” *Advances in Neural Information Processing Systems*, vol. 32, 2019.
- [81] X. Ma, L. Zhao, G. Huang, Z. Wang, Z. Hu, X. Zhu, and K. Gai, “Entire space multi-task model: An effective approach for estimating post-click conversion rate,” in *The 41st International ACM SIGIR Conference on Research & Development in Information Retrieval*, 2018, pp. 1137–1140.
- [82] Z.-F. Gao, P. Liu, W. X. Zhao, Z.-Y. Lu, and J.-R. Wen, “Parameter-efficient mixture-of-experts architecture for pre-trained language models,” *arXiv preprint arXiv:2203.01104*, 2022.
- [83] Y. Li, B. Hui, Z. Yin, M. Yang, F. Huang, and Y. Li, “Pace: Unified multi-modal dialogue pre-training with progressive and compositional experts,” in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2023, pp. 13 402–13 416.
- [84] Y. Zhang, R. Cai, T. Chen, G. Zhang, H. Zhang, P.-Y. Chen, S. Chang, Z. Wang, and S. Liu, “Robust Mixture-of-Expert Training for Convolutional Neural Networks,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 90–101.
- [85] S. Chen, Z. Jie, and L. Ma, “Llava-mole: Sparse mixture of lora experts for mitigating data conflicts in instruction finetuning mllms,” *arXiv preprint arXiv:2401.16160*, 2024.
- [86] T. Chen, Z. Zhang, A. K. JAISWAL, S. Liu, and Z. Wang, “Sparse MoE as the New Dropout: Scaling Dense and Self-Slimmable Transformers,” in *The Eleventh International Conference on Learning Representations*, 2022.
- [87] K. Zhou, J. Yang, C. C. Loy, and Z. Liu, “Learning to prompt for vision-language models,” *International Journal of Computer Vision*, vol. 130, no. 9, pp. 2337–2348, 2022.
- [88] K. Shuster, J. Xu, M. Komeili, D. Ju, E. M. Smith, S. Roller, M. Ung, M. Chen, K. Arora, J. Lane *et al.*, “Blenderbot 3: a deployed conversational agent that continually learns to responsibly engage,” *arXiv preprint arXiv:2208.03188*, 2022.
- [89] M. Li, S. Gururangan, T. Dettmers, M. Lewis, T. Althoff, N. A. Smith, and L. Zettlemoyer, “Branch-Train-Merge: Embarrassingly Parallel Training of Expert Language Models,” in *First Workshop on Interpolation Regularizers and Beyond at NeurIPS 2022*, 2022.
- [90] T. Dao, D. Fu, S. Ermon, A. Rudra, and C. Ré, “Flashattention: Fast and memory-efficient exact attention with io-awareness,” *Advances in Neural Information Processing Systems*, vol. 35, pp. 16 344–16 359, 2022.
- [91] X. L. Li and P. Liang, “Prefix-Tuning: Optimizing Continuous Prompts for Generation,” in *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, 2021, pp. 4582–4597.
- [92] S. Kudugunta, Y. Huang, A. Bapna, M. Krikun, D. Lepikhin, M.-T. Luong, and O. Firat, “Beyond Distillation: Task-level Mixture-of-Experts for Efficient Inference,” in *Findings of the Association for Computational Linguistics: EMNLP 2021*, 2021, pp. 3577–3599.
- [93] Y. Shen, Z. Guo, T. Cai, and Z. Qin, “JetMoE: Reaching Llama2 Performance with 0.1 M Dollars,” *arXiv preprint arXiv:2404.07413*, 2024.
- [94] T. Wei, B. Zhu, L. Zhao, C. Cheng, B. Li, W. Lü, P. Cheng, J. Zhang, X. Zhang, L. Zeng *et al.*, “Skywork-MoE: A Deep Dive into Training Techniques for Mixture-of-Experts Language Models,” *arXiv preprint arXiv:2406.06563*, 2024.
- [95] Z. Ma, J. He, J. Qiu, H. Cao, Y. Wang, Z. Sun, L. Zheng, H. Wang, S. Tang, T. Zheng *et al.*, “BaGuaLu: targeting brain scale pretrained models with over 37 million cores,” in *Proceedings of the 27th ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming*, 2022, pp. 192–204.
- [96] L. Ouyang, J. Wu, X. Jiang, D. Almeida, C. Wainwright, P. Mishkin, C. Zhang, S. Agarwal, K. Slama, A. Ray *et al.*, “Training language models to follow instructions with human feedback,” *Advances in Neural Information Processing Systems*, vol. 35, pp. 27 730–27 744, 2022.
- [97] M. I. Jordan and R. A. Jacobs, “Hierarchical mixtures of experts and the EM algorithm,” *Neural computation*, vol. 6, no. 2, pp. 181–214, 1994.
- [98] A. Q. Jiang, A. Sablayrolles, A. Roux, A. Mensch, B. Savary, C. Bamford, D. S. Chaplot, D. d. l. Casas, E. B. Hanna, F. Bressand *et al.*, “Mixtral of experts,” *arXiv preprint arXiv:2401.04088*, 2024.
- [99] Y. Zheng and D. X. Wang, “A survey of recommender systems with multi-objective optimization,” *Neurocomputing*, vol. 474, pp. 141–153, 2022.
- [100] Y. Bengio, N. Léonard, and A. Courville, “Estimating or propagating gradients through stochastic neurons for conditional computation,” *arXiv preprint arXiv:1308.3432*, 2013.